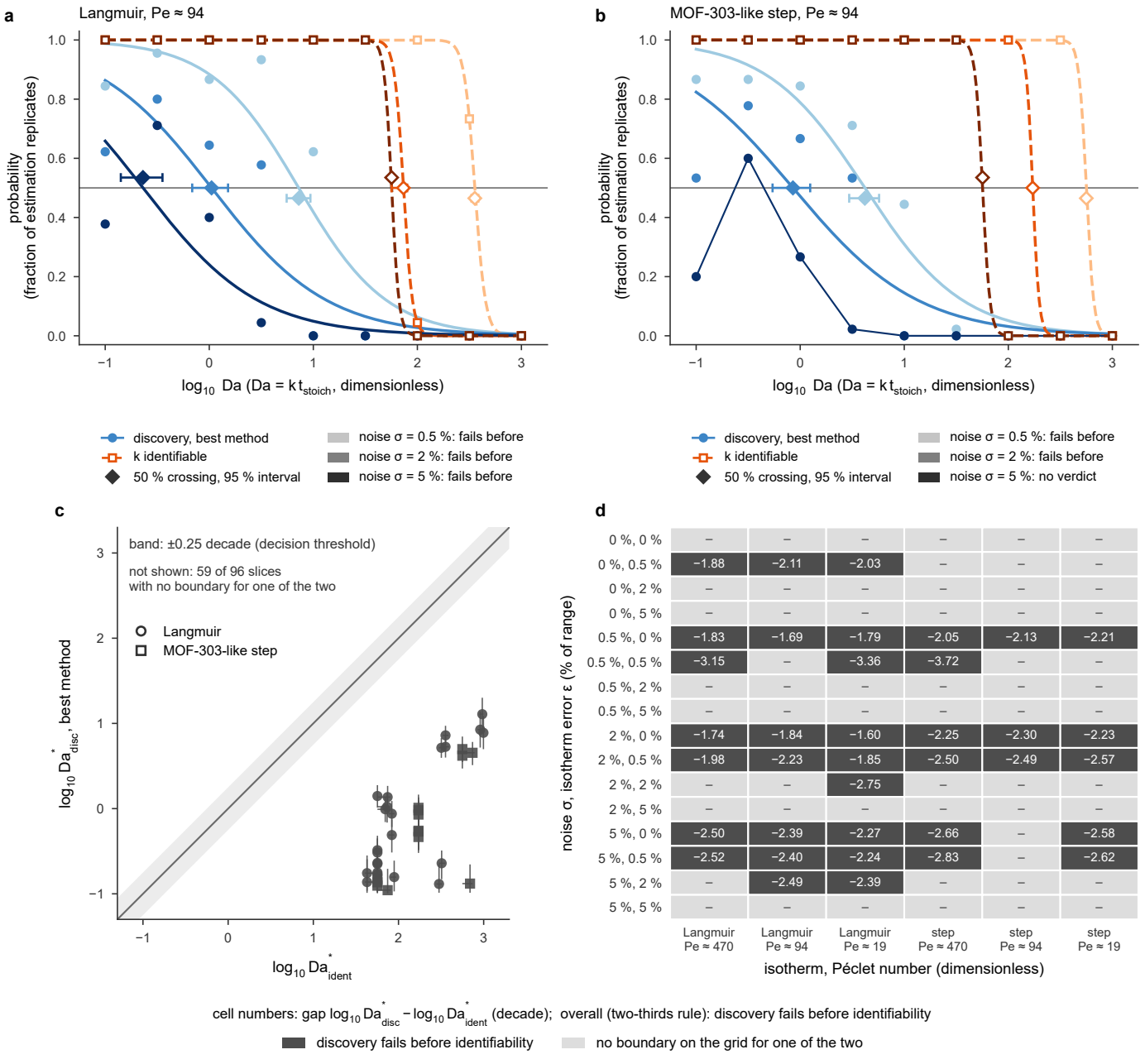

a
Langmuir, Pe ≈ 94
b
MOF-303-like step, Pe ≈ 94
probability
(fraction of estimation replicates)
log10 Da (Da = k tstoich, dimensionless)
discovery, best method
k identifiable
50 % crossing, 95 % interval
noise σ = 0.5 %: fails before
noise σ = 2 %: fails before
noise σ = 5 %: fails before
noise σ = 5 %: no verdict
c
band: ±0.25 decade (decision threshold)
not shown: 59 of 96 slices
with no boundary for one of the two
Langmuir
MOF-303-like step
log10 Da*disc, best method
log10 Da*ident
d
noise σ, isotherm error ε (% of range)
0 %, 0 %
0 %, 0.5 %
0 %, 2 %
0 %, 5 %
0.5 %, 0 %
0.5 %, 0.5 %
0.5 %, 2 %
0.5 %, 5 %
2 %, 0 %
2 %, 0.5 %
2 %, 2 %
2 %, 5 %
5 %, 0 %
5 %, 0.5 %
5 %, 2 %
5 %, 5 %
−1.88 −2.11 −2.03
−1.83 −1.69 −1.79 −2.05 −2.13 −2.21
−3.15 −3.36 −3.72
−1.74 −1.84 −1.60 −2.25 −2.30 −2.23
−1.98 −2.23 −1.85 −2.50 −2.49 −2.57
−2.75
−2.50 −2.39 −2.27 −2.66 −2.58
−2.52 −2.40 −2.24 −2.83 −2.62
−2.49 −2.39
Langmuir Pe ≈ 470
Langmuir Pe ≈ 94
Langmuir Pe ≈ 19
step Pe ≈ 470
step Pe ≈ 94
step Pe ≈ 19
isotherm, Péclet number (dimensionless)
cell numbers: gap log10 Da*disc − log10 Da*ident (decade); overall (two-thirds rule): discovery fails before identifiability
discovery fails before identifiability
no boundary on the grid for one of the two